



Photo: Deyr 2023 floods, Somalia



SAVING
LIVES
CHANGING
LIVES

Somalia El Niño 2026

Flood Exposure & Preparedness Analysis | Deyr Season, October - December 2026

≈800K –
1.2M

initial preparedness
caseload: flood-exposed
and likely to face high
humanitarian needs.

1.7M

people exposed to flood
hazard under the baseline
scenario (~9% of
population)

3.4M – 3.7M

people exposed to flood hazard
under extreme-case scenario
(~18% - ~19% of population)

670K–2.7M

scenario range, from the
moderate to the worst-case
reference season

A very strong El Niño is likely for late 2026. Its strength matches/surpasses the severe 2023 and 1997 seasons Somalia's two worst flood benchmarks. The Indian Ocean Dipole (IOD), the second driver of heavy Deyr rains, has stayed at the edge of its positive threshold (+0.41°C, 9 August) for three weeks and there is high likelihood to tip over and stay positive through the season. The GHACOF74 outlook issued 18 August confirms a wetter-than-normal Deyr, a >90% likelihood of exceeding 400 mm over parts of central-southern Somalia. However, forecasts tell us the season will be wet, but not how severe. This analysis therefore benchmarks four past El Niño seasons, from mild (2015) to worst case (1997) as a severity scale, as rainfall and river levels data come in, they show which past season 2026 is tracking and figures shift within the range of these benchmark seasons.*

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* [Technical Statement from the 74th Greater Horn of Africa Climate Outlook Forum \(GHACOF74\)](#) - ICPAC

Overview & Overall Findings.

WHAT EL NIÑO TYPICALLY MEANS FOR SOMALIA

-  **Heavier Deyr rains (Oct–Dec)**
higher riverine flood risk on the Shabelle and Juba; flash floods beyond mapped plains.
-  **Harvest & asset losses**
flooded cropland and livestock losses cut the Deyr harvest and household income.
-  **Market & access strain**
washed-out roads disrupt supply, prices and access.
-  **Disease & malnutrition**
standing water drives AWD/cholera, malnutrition.
-  **Compounding pressures**
recovering from drought amid displacement, conflict, and Red Sea import disruptions.
-  **Upside of the rains** pasture regeneration, water recharge and rainfed harvest gains beyond flood zones.

Sources: ICPAC / GHACOF outlooks from past El Niño seasons; FEWS NET East Africa El Niño reports; FAO-SWALIM.

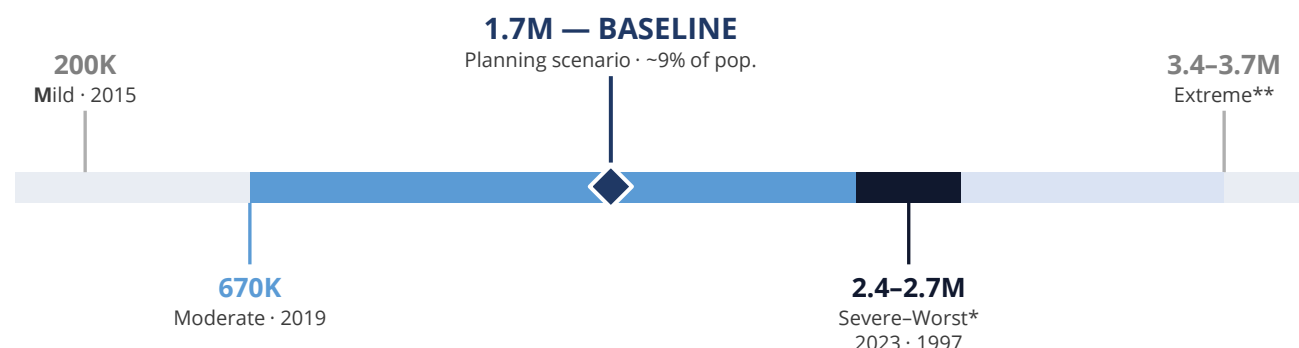
REFERENCE BENCHMARK SEASONS

Season	Tier	Reported Impact	% pop Impacted	Event Description
2015	MILD CASE	90,000 – 150,000 people	~1%	Strong El Niño with neutral IOD produced floods with mild impact.
2019	MODERATE CASE	570,000 people	~4%	Weak El Niño with positive IOD produced moderate flooding.
2023	SEVERE CASE	2,480,000 people	~13.8%	Worst recent flood season; strong El Niño with strong positive IOD.
1997	WORST CASE	+1,000,000 people	+12.8%	Most extreme floods on record; strong El Niño with very strong IOD.
100-yr return flood	EXTREME CASE	n/a	n/a	Compiled from 100 years of flood extents (IGAD return-period layer); 2026 population exposure for this case is calculated via the spatial analysis in subsequent sections.

Where 2026 sits: conditions are tracking above the moderate benchmark, making 2023 the closest analogue.

FLOOD EXPOSURE SCENARIOS FOR DEYR 2026

Population exposure was estimated by applying the 2026 Somalia population estimate of 19.4 million (as of the time of analysis), together with its spatial distribution and density, to each reference flood scenario for Deyr 2026:



Exposure is not the response caseload. These figures estimate people living in flood-hazard areas (riverine and flash floods).

* Due to limitations in the availability and quality of historical data for 1997, the 1997 and 2023 flood events were combined into a single severe/worst-case scenario. While the thresholds and impact estimates for 1997 could not be fully validated, the resulting impact range is broadly consistent with the severity of both events when scaled to Somalia's projected 2026 population. Further details are provided on page 3.

** The 3.2 million figure is based on an OCHA compilation of historical flood extents. The 3.7 million estimate is derived from compilation of the modeled 100-year return period flood extent and 1997 floods. Exposure at this level within a single season has not been observed in Somalia's recorded history and is therefore considered unlikely to occur in one season. However, the current climatic conditions are quite unique, with a potential super El Niño developing. Therefore, an unprecedented event remains within the realm of possibility.

Sources: FAO-SWALIM flood extents (2015, 2019, 2023); IGAD 100-yr return-period flood layer; WorldPop 2026; OCHA & SoDMA reported impact figures (per-season reports listed on p. 5); WFP VAM Somalia flood exposure analysis, Aug 2026.

Analytical approach: from flood exposure to preparedness caseload

ANALYTICAL APPROACH

1

Benchmark: reference from past seasons

Identify which past El Niño seasons 2026 could resemble, from mild to moderate to worst to extreme case.

2

Flood Exposure: Estimate people exposed to floods

2.1

Map riverine flooding

Overlay observed flood extents from FAO-SWALIM (2015, 2019, and 2023) and the IGAD 100-year return-period flood layer with current population data (2026 WorldPop). As the 1997 flood extent dataset was not available for analysis, the 2023 flood extent was used to represent both the severe-case (2023) and worst-case (1997/98) scenarios with triangulation with reported impact*.

2.2

Flag flash-flood areas

Flash flood-prone districts were identified as those exceeding the 75th percentile rainfall threshold over 11 days, or the 90th percentile threshold over 9 days*, during the reference years 1997, 2015, 2019, and 2023, with the 1997 flash flood event and the IGAD 100-year return-period flood layer used to represent the extreme-case scenario. WorldPop 2026 population data was then overlaid to estimate the number of people living in areas likely to have experienced flash flooding

3

Layering current needs: Estimate share of people in Need during Oct – Dec 26***

Compute people in need of assistance by estimating district IPC 3+ shares, based on average shifts observed in 2019 and 2023 per district.

4

Estimated preparedness caseload: flood exposed & food-insecure population

In each district multiply people exposed to floods by the % people in need calculated from previous step.

5

Track in real time

Monitor live climate and river indicators; adjust planning to the tracking scenario.

TRIANGULATING THE EVIDENCE

A · Reported impact figures

Officially reported “people affected” in the four benchmark seasons, expressed as a share of the population at the time (1.1%–13.8%).

X

B · Modelled flood exposure

Riverine: observed FAO-SWALIM flood extents (2015, 2019, 2023) for the mild-severe scenarios; the IGAD 100-year return-period layer bounds the extreme case. Flash flood: extreme-rainfall zones. Population, WorldPop 2026 applied to both hazard layers.



Reconciled on % of population affected.

Where modelled and reported rates converged (~5% margin), the reported rate was retained; where they diverged, the higher rate (or more confident rate) was adopted as a precautionary basis, then applied to current population.

* Since the spatial footprint of the 1997 floods was not available for direct analysis, estimates were derived using reported impacts. However, the resulting exposure estimates were lower than those for the 2023 floods, despite the 1997/98 event being considered more severe. This discrepancy is likely driven by differences in population distribution and exposure patterns. To account for this uncertainty, the 2023 and 1997 estimates were combined into a single severe-to-worst-case range. The upper bound is expected to capture the vast majority of plausible impacts, although a limited degree of residual uncertainty may remain.

** Flash flood methodology thresholds were not formally calibrated; however, the selected ones showed close alignment with reported flood impacts and observed affected areas.

*** Impacts will extend beyond increased humanitarian needs. Affected populations may also experience displacement, damage to housing, infrastructure, losses of agricultural assets and production. Quantifying these household-level impacts is beyond the scope of the present analysis and will be assessed through post-shock assessments and other relevant sectoral evaluations

FROM EXPOSURE TO CASELOAD

District flood-exposed population

scenario hazard layer ×
WorldPop 2026; district mid-
points (lower + upper) ÷ 2

× Estimated share of people in Need

per district, average shift (Δ)
observed in reference year (2019
& 2023)

= flood-exposed & high humanitarian needs

Flood-exposed population
adjusted for the share of people
likely to experience high levels of
need, aggregated across all
districts.

→ Estimated preparedness caseload

These estimates are scenario-based projections intended to inform preparedness planning and should not be interpreted as IPC classifications or empirical needs assessments. The analysis assumes a similar pattern to that observed in 2019 and 2023, while not controlling for other compounding factors that may influence outcomes. Comprehensive food security needs estimates will be available through the IPC report expected in late September 2026. In the meantime, this preliminary analysis aims to isolate and better understand the potential impact of El Niño-related flooding on affected populations and estimate minimum level for preparedness.

Sources: [FAO-SWALIM flood extents](#) (2015, 2019, 2023); [IGAD 100-yr return-period flood layer](#); [CHIRPS](#) rainfall (flash-flood percentile thresholds); [WorldPop 2026](#); [IPC Somalia](#) (Apr–Jun 2026 baseline); WFP VAM calculations.

Seasonal climate outlook and pre-season indicators, Deyr 2026

PRE-SEASON INDICATOR DASHBOARD · AS OF 18 AUGUST 2026

+2.1°C

Niño 3.4 SST anomaly (weekly)

El Niño strength; sustains late-season rain. Already above the 2023 peak (+2.0°C) and trending toward super El Niño strength.

SUPER EL NIÑO RANGE IN REACH

+0.41°C

Dipole Mode Index (IOD)

Primary driver of East African OND rainfall. Third consecutive week at the +0.4°C threshold; event not yet formally established (BoM, 9 Aug).

AT +IOD THRESHOLD · RISING

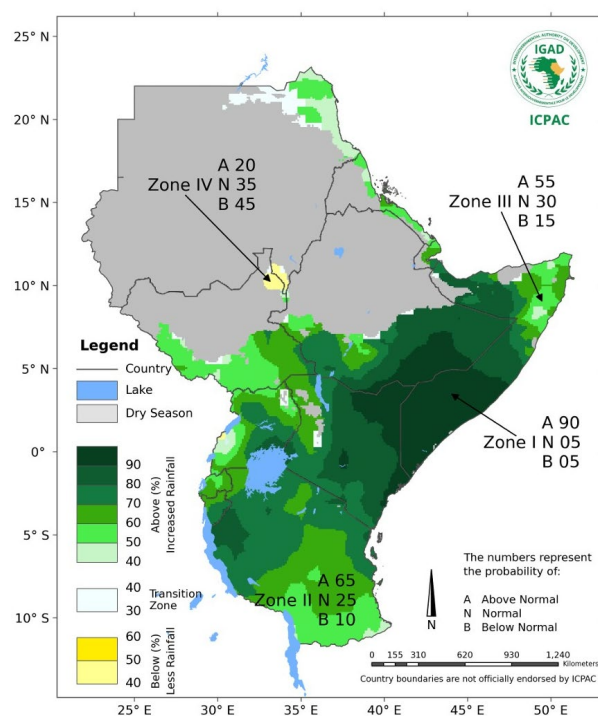
>90%

OND seasonal forecast · GHACOF74

Likelihood of exceeding 400 mm over central-southern Somalia. Wetter-than-normal OND confirmed 18 Aug.

CONFIRMED WET SIGNAL

GHACOF / C3S OND 2026 SEASONAL FORECAST



Zone I (dark green): above-normal rainfall is the most likely category at 90%; normal and below-normal are 5% each.

Zone II (green): above-normal is also most likely; for the most widespread shade, 65% above-normal vs 25% normal and 10% below.

Zone III (light green): above-normal most likely at 55%; normal 30%, below-normal 15%.

Zone IV (yellow): below-normal (drier) most likely at 45%; near-normal 35%, above-normal 20%.

Somalia's riverine south sits in Zones I–II — the highest wet-signal probabilities of the region.

The forecast identified 1997 and 2023 as analogue years, both of which Somalia experienced significant flood events and associated impacts.

Green = above-normal (wet) · grey = normal · yellow = below-normal

HOW 2026 COMPARES ON MAIN DETERMINING INDICATORS

Indicator (OND)	2015 mild	2019 moderate	2023 severe	1997 worst	2026 to date
Niño 3.4 SST anomaly	~+2.4°C	~+0.5°C	~+2.0°C	~+2.4°C	+2.1°C · strengthening
Dipole Mode Index	~+0.5°C	~+2.1°C	~+1.5°C	strong +	+0.41°C · at +IOD threshold – projected to rise
Ethiopian highlands rainfall	100–150%	150–200%	200–300%	300%+ locally	wetter-than-normal OND (GHACOF74)

Sources: [IRI/NOAA CPC](#) (Niño 3.4 weekly anomaly, current and historical); [BoM Climate Driver Update](#) (Dipole Mode Index); [ICPAC GHACOF74](#) / C3S (OND 2026 seasonal forecast and zone probabilities); [CHIRPS](#) (Ethiopian highlands OND rainfall, historical seasons).

Reference El Niño seasons: seasonal rainfall and reported impacts

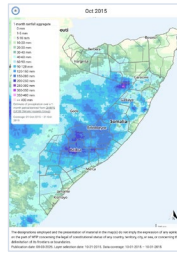
Monthly Deyr rainfall (October–December) shown for each season, with the national average accumulation per month (CHIRPS).

MILD

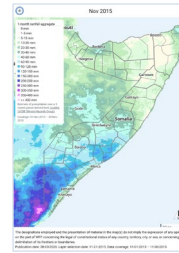
2015

Very strong El Niño · weak IOD

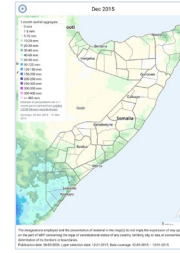
Flooding at the mild end of the range



OCT · 78 mm avg



NOV · 46 mm avg



DEC · 9 mm avg

REPORTED PEOPLE
AFFECTED

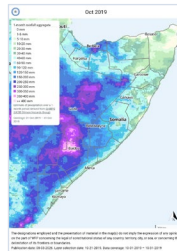
~145K affected | ~1%
~42K displaced

MODERATE

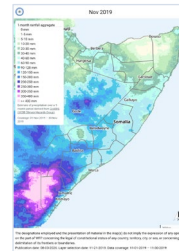
2019

Record positive IOD · near-neutral ENSO

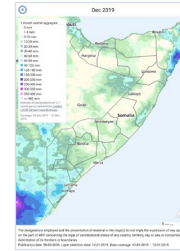
Widespread but less severe flooding



OCT · 111 mm avg



NOV · 88 mm avg



DEC · 21 mm avg

REPORTED PEOPLE
AFFECTED

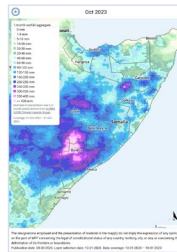
~570K affected | ~4%
~370K displaced

SEVERE

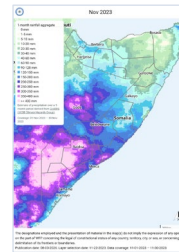
2023

Strong El Niño · strong positive IOD

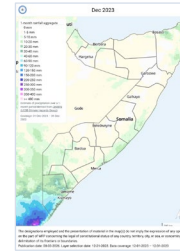
Among the heaviest rains on record



OCT · 120 mm avg



NOV · 154 mm avg



DEC · 6.3 mm avg

REPORTED PEOPLE
AFFECTED

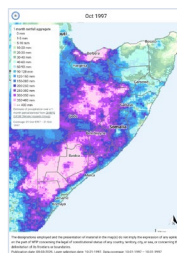
~2.48M affected | ~13.8%
~1.2M displaced

WORST CASE

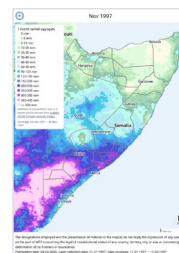
1997

Very strong El Niño · strong IOD

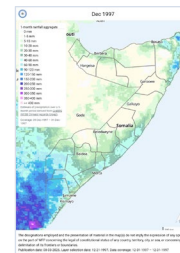
Exceptional rainfall; severe flooding



OCT · 215 mm avg



NOV · 115 mm avg



DEC · 26 mm avg

REPORTED PEOPLE
AFFECTED

+1.0M affected | +12.8%
+2,000 death

SEASONAL FLOOD-RISK CALENDAR

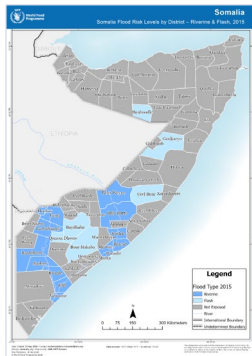
FLOOD-RISK WINDOW

	Aug '26	Sep '26	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27
South / Central Somalia	Wetter · flood risk		Wetter · HIGH FLOOD RISK			Jilaal (dry season)		
Northern Somalia (Somaliland)	Near average		Wetter · flood risk			Jilaal (dry season)		
Northern Somalia (Puntland)	Near average		Wetter · flood risk			Jilaal (dry season)		

Sources: CHIRPS monthly rainfall estimates — Climate Hazards Center, UC Santa Barbara (chc.ucsb.edu/data/chirps); national average accumulation aggregated by WFP VAM. Reported affected and displaced figures: Inter-Agency Flood Response Update (Mar 1998); OCHA Flash Update No. 2 (Nov 2015); OCHA Flood Response Snapshot (Dec 2019); OCHA & SoDMA Deyr Floods SitRep No. 5 (Dec 2023).

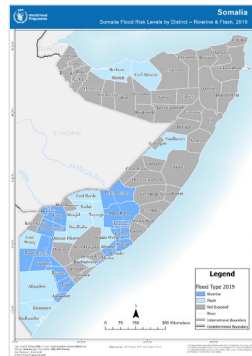
Flood exposure by scenario and district-level risk classification

MILD CASE



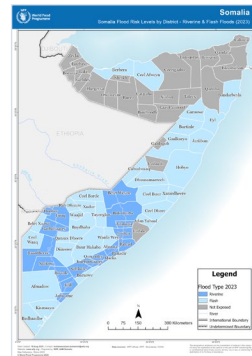
200K people
17 districts · 2015 extent

MODERATE CASE



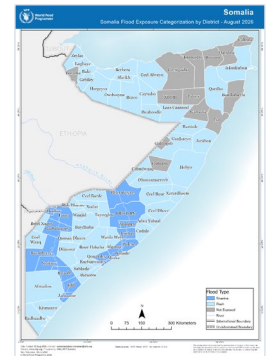
670K people
31 districts · 2019 extent

SEVERE CASE



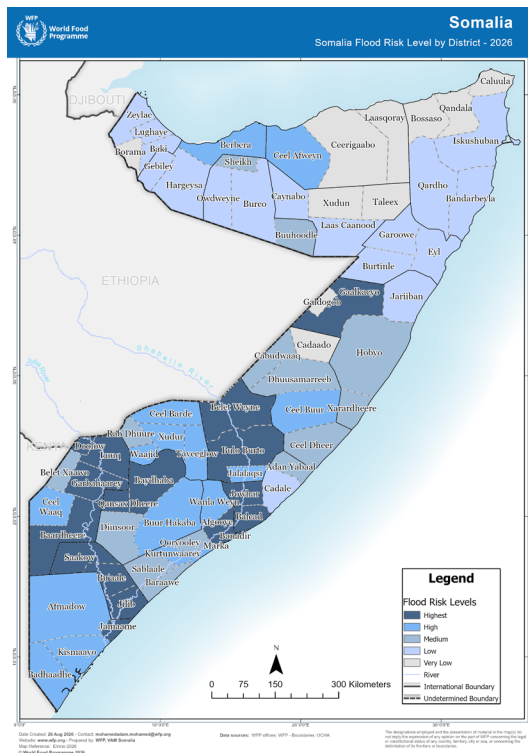
2.7M people
49 districts · 2023 extent

EXTREME CASE

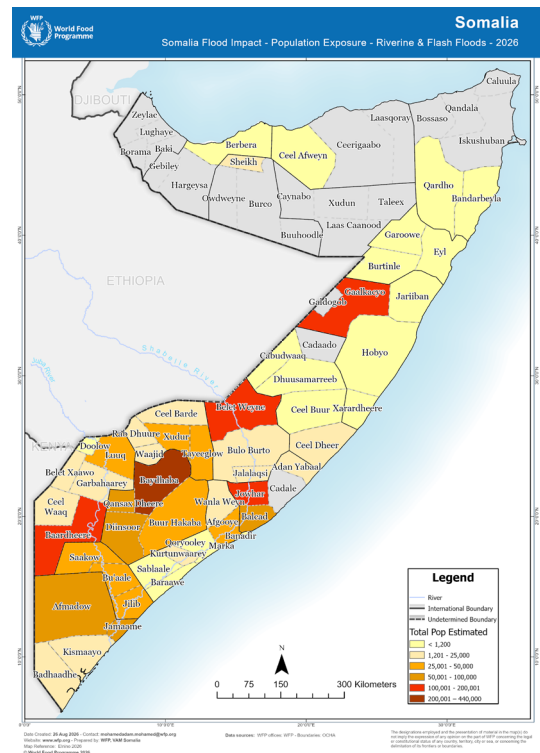


3.7M people
57 districts · 100-yr layer

DISTRICT FLOOD-RISK CLASSIFICATION



POPULATION AT RISK OF FLOOD EXPOSURE · DEYR 2026



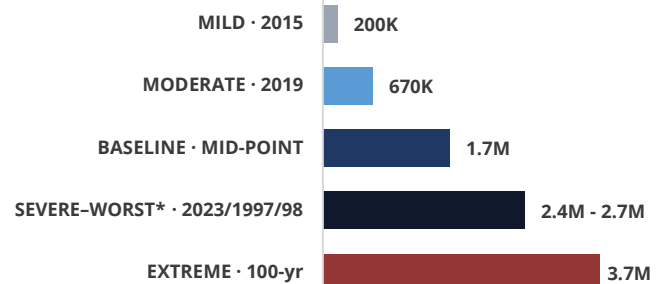
Flash-flood exposure draws on historical flood reporting and satellite detection, both can miss fast, localized events, so some high-risk districts show low or zero exposed population. **Low figures in high-risk districts are not evidence of safety** read them alongside contextual knowledge.

DISTRICTS BY RISK LEVEL

Risk levels reflect the four spatial analysis periods: districts flooded in all four are classed highest; fewer exposures step the class down.

RISK LEVEL	N ^o OF DISTRICTS
Highest	17
High	16
Medium	14
Low	15
Very Low	12

EXPOSURE BY SCENARIO · PEOPLE



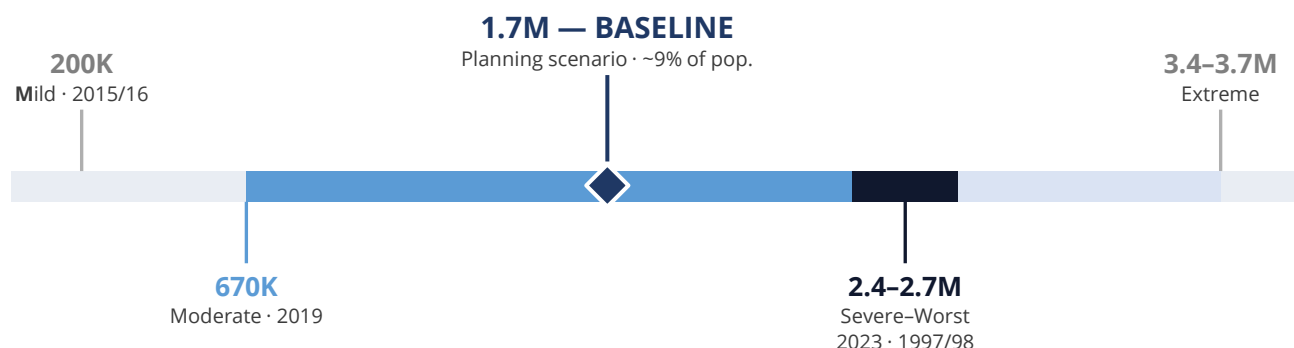
* No 1997 flood extent exists to map; ≈2.4M applies the reported 12.8% affected share (the only 1997/98 data available) to the 2026 population of 19.4M. The severe (2023) and worst-case (1997) results are combined in the single 2.4–2.7M band.

Sources: [IGAD 100-year flood layer](#); [FAO-SWALIM flood extents](#) (2015, 2019, 2023); [WorldPop 2026](#); [CHIRPS flash-flood thresholds](#); WFP VAM calculations.

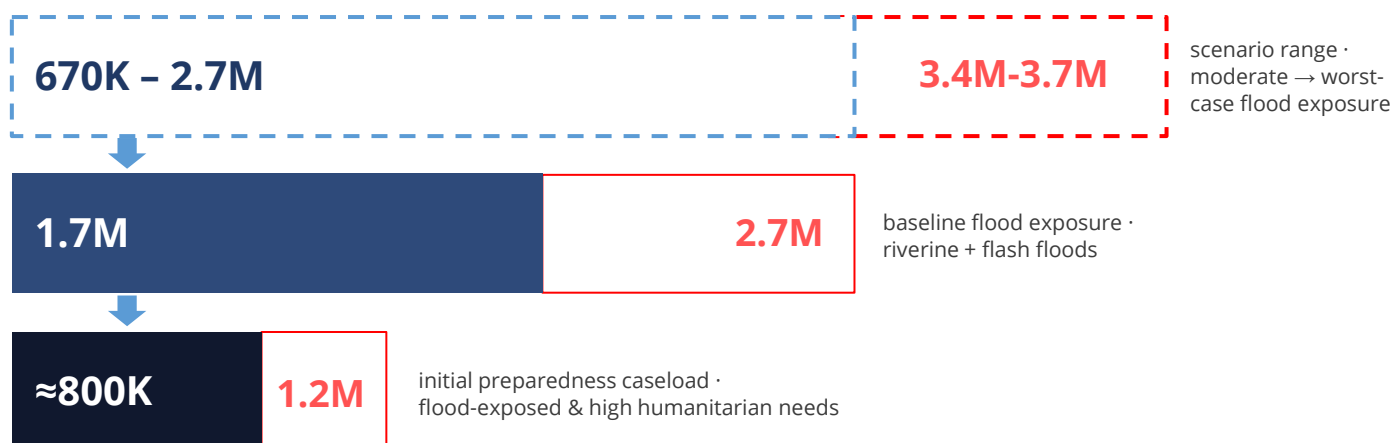
Regional flood exposure and preparedness figures, October–December 2026

FLOOD EXPOSURE SCENARIOS FOR DEYR 2026

Deyr 2026 is projected to track above the moderate case toward the 2023 severe benchmark (page 4), more severe than 2019. The planning baseline therefore starts at the mid-point between the moderate and severe-worst cases (1.7M), with flexibility to scale up plan to the full 2.7M severe-worst exposure.



FROM SCENARIO RANGE TO PLANNING CASELOAD



REGIONAL PREPAREDNESS FIGURES · OCT-DEC 2026 *

flood-exposed & high humanitarian needs

Region	Flood Exposed Lower bound	Flood Exposed Upper bound	Flood Exposed Mid-point	Estimated Preparedness Caseload**	Estimated Preparedness Caseload Upper Bound
Bay	125K	982K	553K	276K	489K
Hiraan	200K	197K	199K	92K	90K
Middle Shabelle	47K	320K	183K	58K	102K
Lower Juba	196K	165K	180K	66K	74K
Gedo	21K	327K	174K	58K	109K
Mudug	-	339K	169K	62K	124K
Middle Juba	14K	172K	93K	35K	64K
Bakool	29K	104K	67K	19K	29K
Lower Shabelle	2K	106K	54K	21K	42K
Banadir	25K	17K	21K	3K	2K
Togdheer	14K	-	7K	847	-
Galguduud	-	4K	2K	826	2K
Nugaal	-	2K	1K	435	869
Woqooyi Galbeed	609	1K	988	229	316
Sanaag	1K	164	687	254	61
Bari	-	516	258	111	223
Awdal	-	-	-	-	-
Sool	-	-	-	-	-
National total (18 regions)	~670K	2.7M	~1.8M	~800K	~1.2M

*The figures presented here are indicative and are based on desk review and analysis. As more comprehensive assessments and data from multiple sources become available, these estimates should be further triangulated and validated to ensure accuracy and completeness.

**The figures presented here should not be confused with IPC classifications or other established measures of food insecurity and humanitarian needs. They are indicative estimates intended to provide a broad indication for preparedness.

Sources: FAO-SWALIM flood extents; WorldPop 2026; IPC Somalia (Apr–Jun 2026); CHIRPS flash-flood thresholds; WFP VAM calculations.

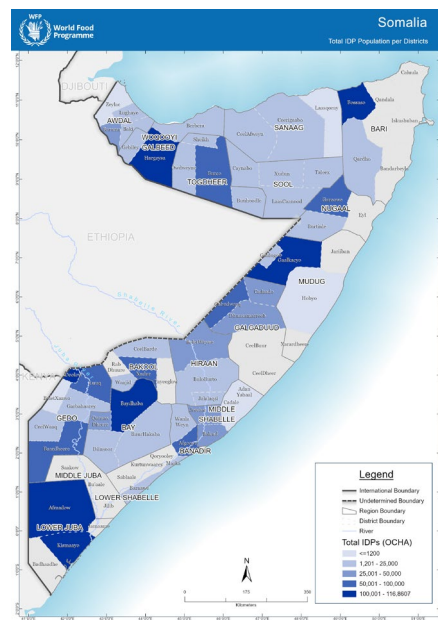
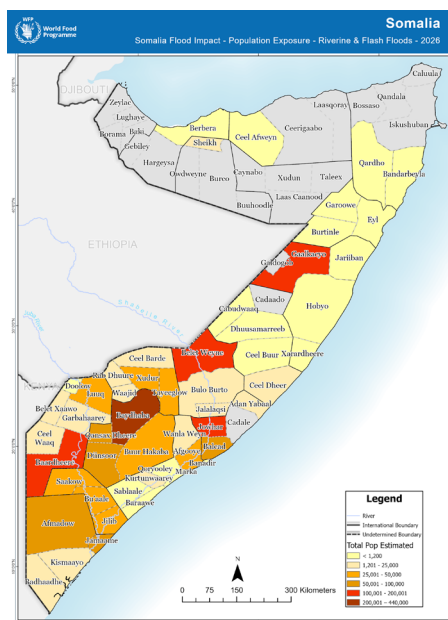
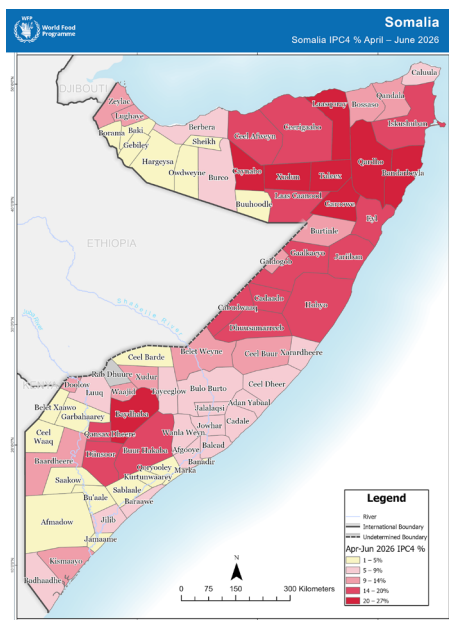
Sub-district analysis, food insecurity, flood exposure and displacement

COMPOUNDING EFFECTS

IPC PHASE 4 % ·
APR-JUN 2026

FLOOD EXPOSURE ·
MID-POINTS

INTERNALLY DISPLACED
PEOPLE · OCHA 2026



The absence of reported floods does not necessarily mean that flooding has not occurred or will not occur in the future. The data are based on a compilation of historical flood records, and certain events, particularly localized or flash floods, may be underreported or missed. Therefore, areas with no recorded floods should be interpreted with caution and along side local knowledge.

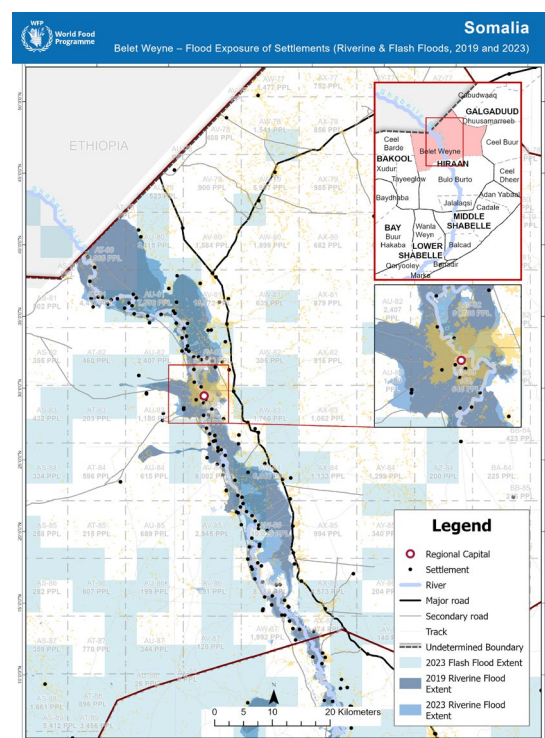
There is a significant overlap between areas currently facing high acute IPC Phase 4 needs, high levels of displacement, and high exposure to flooding in Deyr 2026.

SUB-DISTRICT IDENTIFICATION

District-level maps can be used operationally to identify the most at-risk locations by overlaying flood extent data from 2019 to 2023. Villages or settlements that were affected by flooding in multiple years can be classified as the highest-risk settlements, while those affected in only one year should still be considered high risk, particularly given the forecasted intensity of upcoming rainfall. Sub-district level analysis can be done on a case-by-case basis.

Belet Weyne District	
Flood Exposed Lower bound	160,000 People
Flood Exposed Upper bound	180,000 People
Flood Exposed Mid-point	170,000 People
Estimated Preparedness Caseload	85,000 People
Estimated Preparedness Caseload Upper Bound	100,000 People
Settlements in High Risk of Riverine Floods	198 Settlements
Settlements in Highest Risk of Riverine Floods	85 Settlements

BELET WEYNE DISTRICT



The figures presented here are based on preliminary analysis. As more comprehensive assessments and data from multiple sources become available, these estimates should be further triangulated and validated to ensure accuracy and completeness.

Sources: IPC Somalia (Apr-Jun 2026); OCHA IDP figures (2026); settlements from OCHA 2022 compilation, rectified with Meta building footprints; flood extents from FAO-SWALIM; flash floods from CHIRPS rainfall thresholds (see Methodology, p. 3); administrative boundaries from OCHA.

Planning implications and monitoring

≈800,000 – 1,200,000 is the initial preparedness caseload for operational planning, while retaining flexibility to scale up or down within the 670K–2.7M exposure range depending on how the Deyr evolves and on observed impacts. El Niño strength is already comparable to the 2023 and 1997 analogues; if the positive IOD consolidates, planning should shift toward the severe and worst-case scenarios pre-positioning and pipeline decisions should anticipate the upper bound of the range.

INDICATORS TO MONITOR

Indicator	What to watch	Key source
El Niño strength	Weekly Niño 3.4 anomaly; whether the very strong signal holds into OND	IRI / NOAA CPC
Indian Ocean Dipole	DMI holding above the positive-IOD threshold	BoM Climate Driver Update
Seasonal forecasts	OND issuances for the Ethiopian highlands and southern Somalia	Weekly forecast CHIRPS and NOAA
River levels & flood watch	Juba and Shabelle levels vs. moderate / high flood-risk thresholds	FAO-SWALIM flood watch & Ministry of Water and Energy
Observed rainfall & impacts	Live tracking vs. mild / severe / worst-case benchmarks; reported flooding	WFP PRISM · CHIRPS · OCHA/SoDMA

OPERATIONAL IMPLICATIONS

Preparedness based on the baseline

Use the ≈800K-1,200,000 caseload and the district-level exposure ranges to guide pre-positioning and partner coordination before the window opens in October.

Watch the corridors of risk

Riverine districts along the Juba and Shabelle, Bay, Hiraaan and the Shabelle/Juba regions hold ~70% of flood exposed with high food insecurity.

Re-analyse as indicators land

Monitoring the indicators and re-run the analysis based on observed rainfall and impact. Consider river breakages into the analysis at sub-district level.

RESOURCES

WFP HungerMap LIVE
live food security monitoring

PRISM
climate risk monitoring

IPC
acute food insecurity analyses

WFP DataViz
food security & market data

FAO SWALIM
flood watch; water & land info

FEWS NET · ICPAC
outlooks & regional climate



Scan for the live
Somalia HungerMap

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